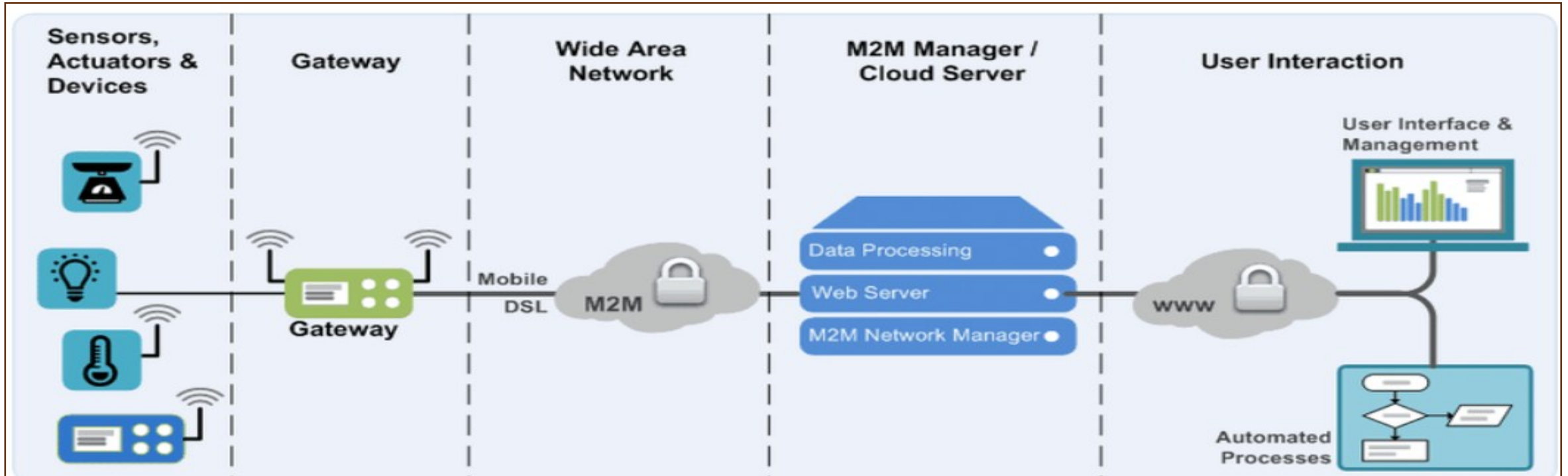
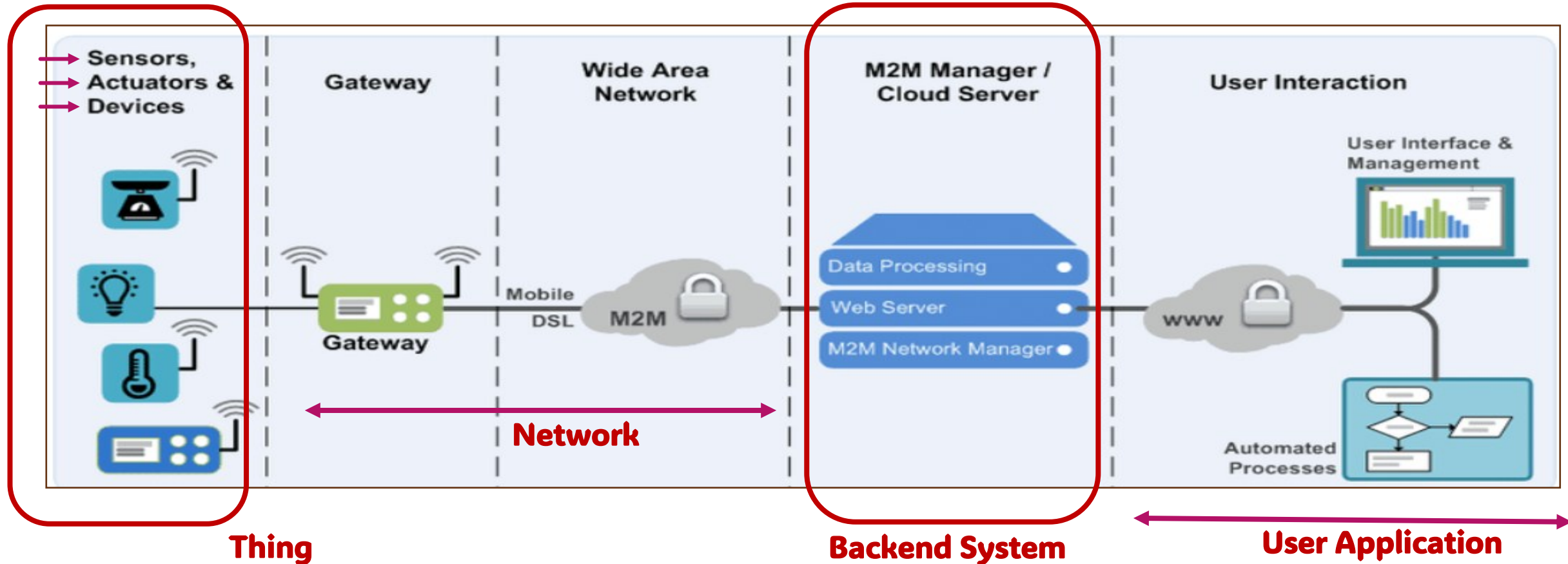


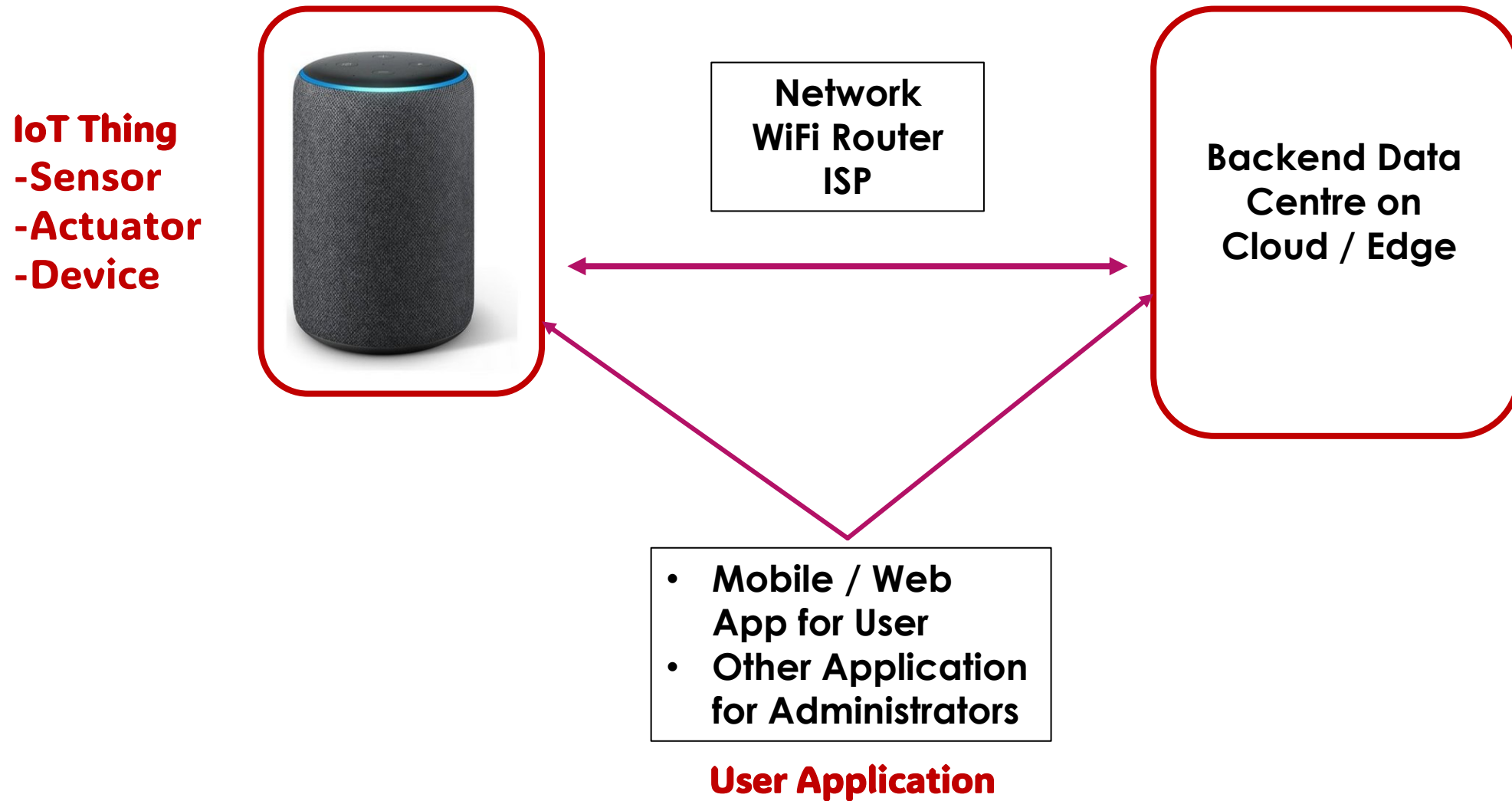
IoT Architecture and Challenges



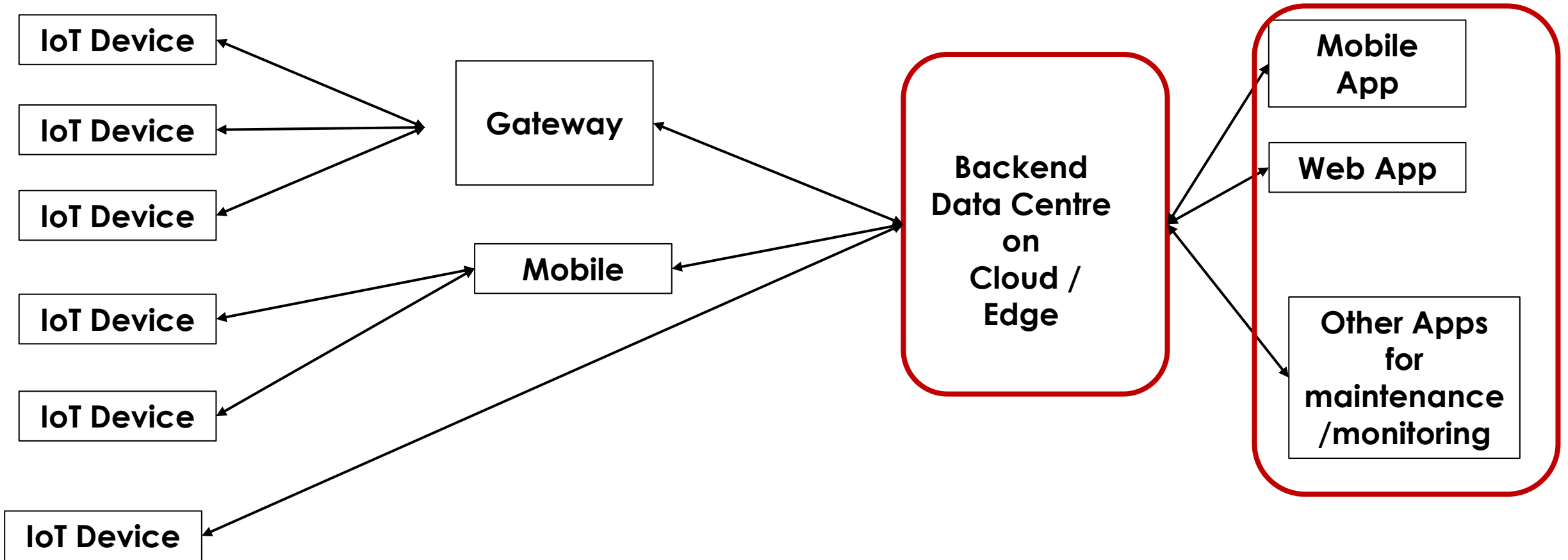
IoT Architecture and Challenges



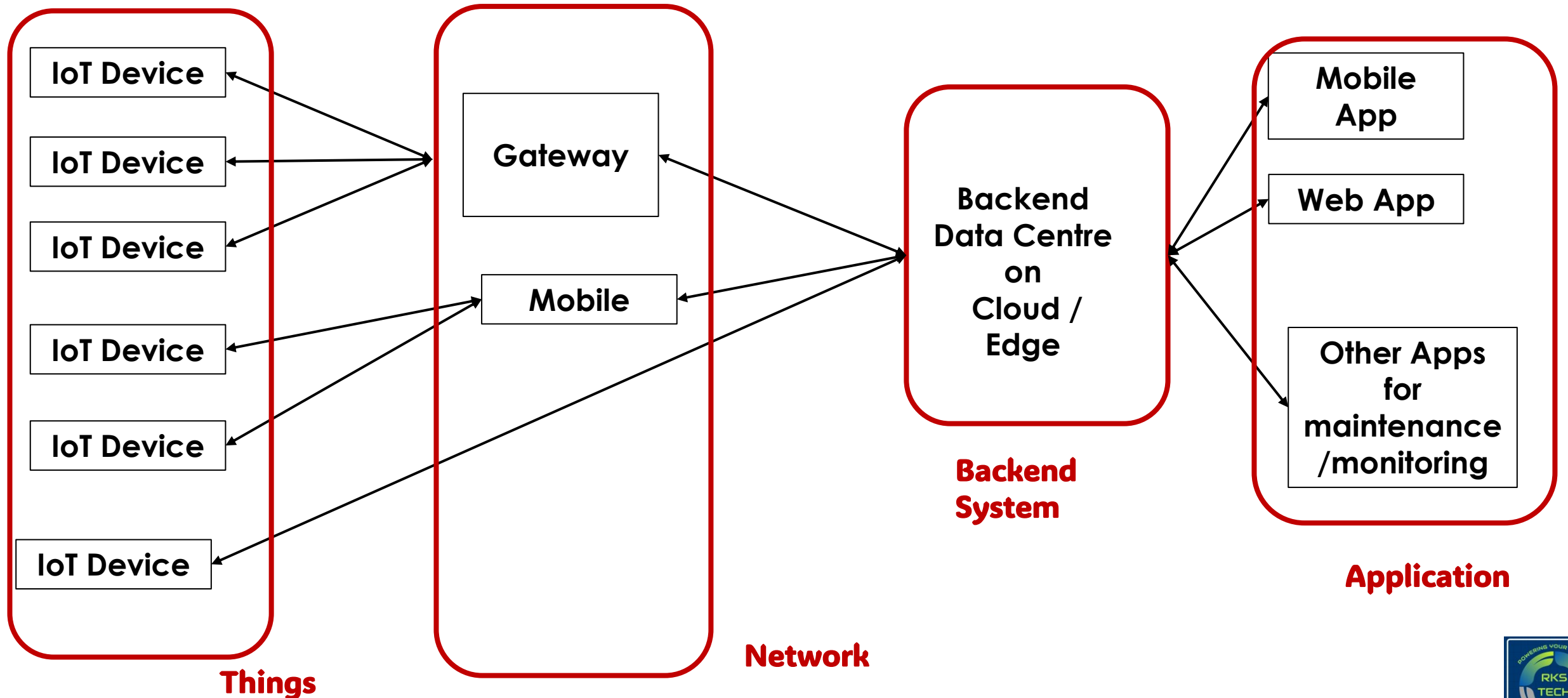
IoT Architecture and Challenges



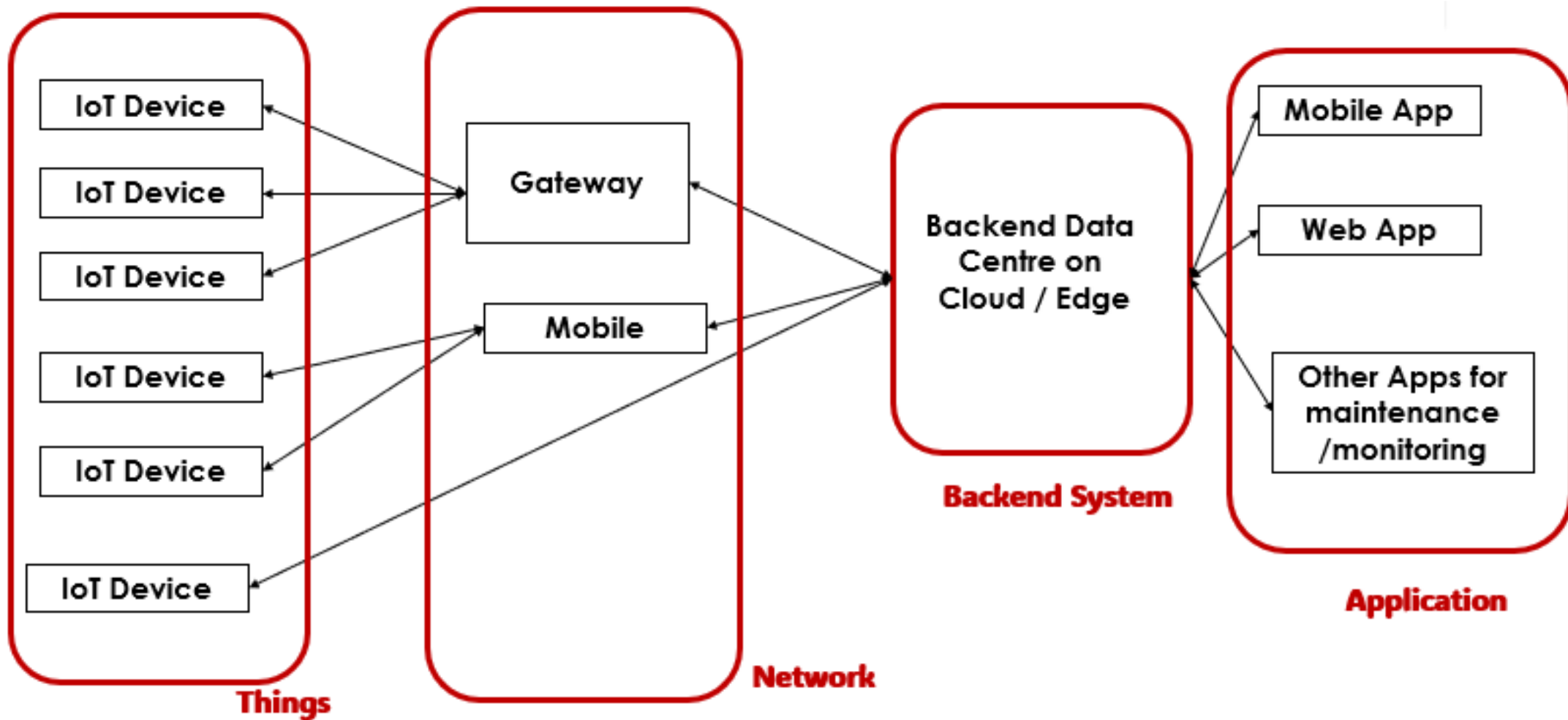
IoT Architecture and Challenges



IoT Architecture and Challenges



IoT Architecture and Challenges



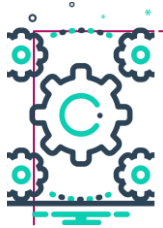
IoT Architecture and Challenges



**No single classification
- Many variants**



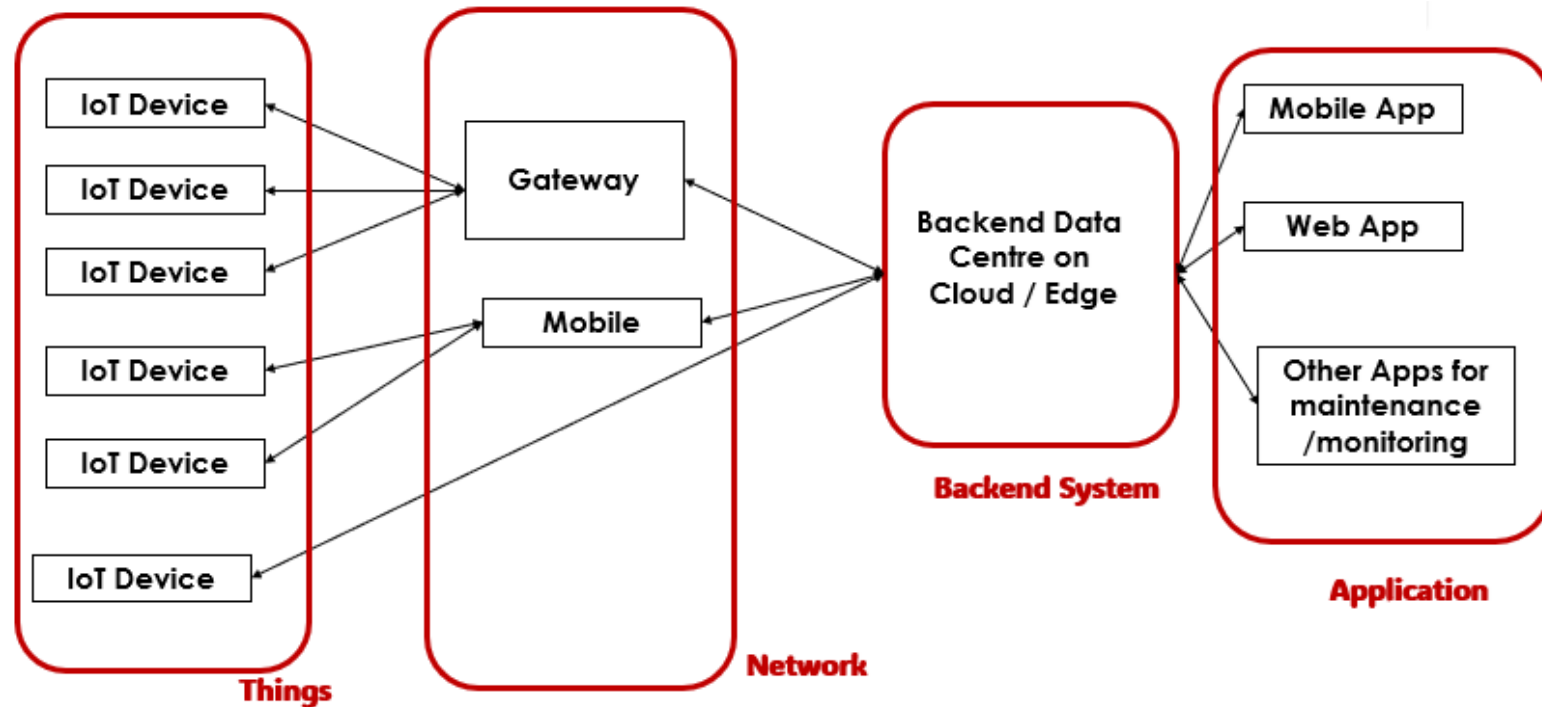
**Connecting
Heterogeneous
system**



**Interoperability &
Standardization**



**Multiple layers of
processing - Device,
Cloud, Fog/Mist/Edge**



IoT Architecture and Challenges



Security & Privacy Risks



Scalability & Network Congestion

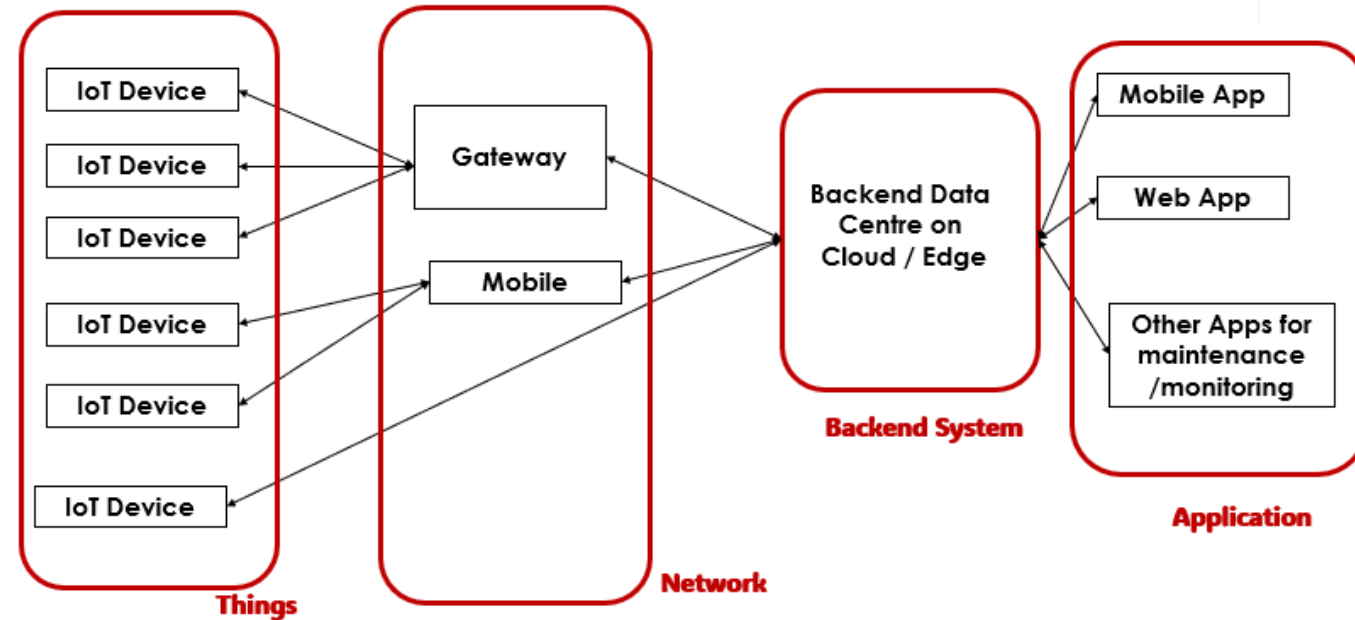


Long battery life

Power Consumption & Battery Life



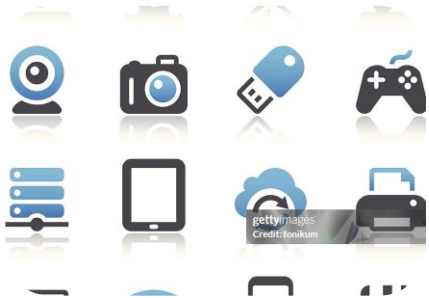
Data Management & Storage



Interoperability & Standardization



Interoperability - ability of different IoT devices, platforms, and systems to seamlessly communicate, exchange data, and work together effectively, regardless of their manufacturer, communication protocols, or underlying technology.



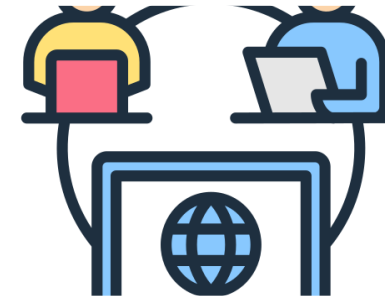
Device Interoperability



Network Interoperability



Data Interoperability



Platform Interoperability



Security Interoperability

IoT Architectural Components



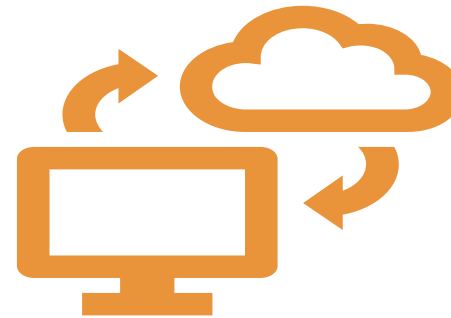
IoT Thing

- Sensors
- Actuators
- Devices



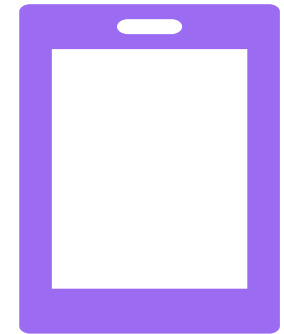
Communication

- WiFi, Gateways
- Mobile/GPRS.4G/5G,
- Satellite,



Computing

- Devices
- Cloud
- Edge/ Fog / Mist



Application

- Users
- Administration / Reporting
- Monitoring / Tracking

IoT Architectural Components

IoT is about interconnected smart devices.

IoT is about collecting data, communicating data, and processing data.

IoT is not a single technology—it is an ecosystem of multiple technologies working together to deliver real business value.

In the upcoming modules we will take deeper dive in

- IOT devices including sensors, and actuators,
- Networking and communication protocols,
- cloud computing and data processing.
- Integrating all there to develop end to end solution